EV Battery Course — Module 9: Charging Technology and Battery Performance

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The charger offers power; the battery decides how much it will accept. On the 60 kWh pack from Module 6, a 100 kW DC charger takes about 35 minutes for 10–80%, a 50 kW charger about 55 minutes, and a 7.4 kW home wallbox about 8 hours for 10–100%. This module explains why, covering AC and DC charging, charging curves, lithium plating, regen, V2X, swapping and standards.

1. Learning objectives

- Distinguish AC and DC charging, and onboard vs offboard chargers.
- Explain CC-CV charging and why charging tapers.
- Relate charging power, C-rate, voltage architecture and temperature.
- Explain lithium plating and how BMS charging limits prevent it.
- Calculate realistic charging times including losses and taper.
- Explain regenerative braking, V2L/V2H/V2G and battery swapping.
- Name the main charging connectors and standards, especially in India.

2. Fundamentals: AC vs DC

	AC charging	DC fast charging
Where AC is converted to DC	Onboard charger (OBC) inside the car	Offboard charger (the charging station)
Typical power	3.3 kW (15 A socket), 7.4 kW (single-phase wallbox), 11–22 kW (three-phase)	25–60 kW (common in India), 100–350 kW (highway hubs)
Limited by	OBC rating, supply	Station rating, vehicle acceptance, pack voltage and temperature
Typical use	Home, workplace, overnight	Highway, fleets, top-ups

The OBC is small and limited by cost, mass and heat inside the car. DC chargers put large power electronics in the station and talk directly to the BMS.

3. Detailed technical explanation

CC-CV charging. In the **constant current (CC)** phase, the charger delivers a fixed current; cell voltage rises. When the highest cell reaches its maximum voltage (e.g. 3.65 V LFP, 4.2 V NMC), the **constant voltage (CV)** phase holds voltage while current falls. CV is slow but fills the last few percent safely.

EV fast-charging curves are more complex. The BMS requests a current that changes continuously with SOC, temperature and cell voltage (a "multi-stage" or "step" curve), not a simple CC-CV.

Why charging tapers.

1. As SOC rises, the anode's ability to absorb lithium quickly falls, raising plating risk.
2. Cell voltage plus I × R approaches the maximum voltage, so current must reduce.
3. Heat builds up; the BMS lowers current to keep the hottest cell below its limit.
4. Imbalance: the highest cell hits limits first.

Lithium plating. When lithium ions arrive at graphite faster than they can intercalate (high current, low temperature, high SOC), they deposit as metallic lithium on the surface. Consequences: permanent capacity loss, more SEI, and dendrites that can pierce the separator. The BMS's charging map (allowed current vs SOC and temperature) is built from cell tests to stay below the plating threshold.

Voltage architecture. Power = V × I. Cable and connector current limits (often 200–500 A for CCS2) cap power: at 350 V, 400 A = 140 kW; at 750 V, 400 A = 300 kW. 800 V cars reach high powers without huge currents.

Charging losses. Station power electronics (about 93–96% efficient), cable and connector resistance, battery I^2R, and the energy used to cool the pack. AC charging adds OBC losses (about 90–95%); at low power, fixed loads (BMS, pumps) make efficiency worse. Overall grid-to-battery efficiency is commonly 85–95%.

4. Diagram: a typical DC charging curve

```
xychart-beta
  title "Illustrative 100 kW DC charging curve, 60 kWh LFP pack"
  x-axis "State of charge (%)" [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
  y-axis "Charging power (kW)" 0 --> 110
  line [100, 100, 100, 100, 70, 70, 45, 30, 15, 5]
```

The flat region is limited by the charger or vehicle current limit; after about 50% the BMS reduces power; above 80% it falls steeply.

5. Step-by-step: a DC fast-charging session

1. Plug in; the vehicle and station perform a handshake (CCS2: ISO 15118 / DIN 70121 over power-line communication; Bharat DC-001, GB/T and CHAdeMO use CAN).
2. Insulation check and contactor closure; BMS reports maximum voltage and current.
3. Charger ramps up current; BMS requests changing current every second or so.
4. Thermal system cools the pack; BMS tapers current with SOC and temperature.
5. At target SOC or user stop, current ramps down, contactors open, session ends and billing is recorded.

6. Engineering equations

$$P_{charge} = V_{pack} \times I \qquad \text{C-rate} = \frac{I}{C_{pack}} \approx \frac{P\ (\text{kW})}{E\ (\text{kWh})}$$

$$t = \sum_i \frac{\Delta E_i}{P_i \times \eta_i}$$

Split the SOC window into bands i, each with its energy ΔE_i (kWh), average charger power P_i (kW) and efficiency η_i (charger output reaching the cells).

Regenerative braking energy:

$$E_{regen} = \eta_{regen} \times \tfrac{1}{2} m (v_1^2 - v_2^2)$$

7. Worked examples on the 60 kWh pack

Pack: 61.44 kWh gross. Each 10% of SOC = 6.144 kWh. Assume 95% of charger output reaches the cells in DC charging and 90% for AC.

Example 1: 100 kW charger, 10–80%. Use the curve above.

SOC band	Energy	Power	Time
10–50%	24.58 kWh	100 kW × 0.95	15.5 min
50–70%	12.29 kWh	70 kW × 0.95	11.1 min
70–80%	6.14 kWh	45 kW × 0.95	8.6 min
Total	43.0 kWh	—	**35.2 min**

A naive estimate (43.0 kWh ÷ 100 kW = 26 min) underestimates by about a third.

Example 2: 50 kW charger, 10–80%. The car can take more than 50 kW up to 70%, so the charger limits: 36.86 kWh ÷ (50 × 0.95) = 46.6 min; then 70–80% at 45 kW: 8.6 min → **55.2 min**.

Example 3: AC charging, 10–100%. 55.3 kWh into the cells.

- 7.4 kW wallbox: 55.3 ÷ (7.4 × 0.9) = **8.3 h**
- 3.3 kW (15 A socket): 55.3 ÷ (3.3 × 0.9) = **18.6 h**

(The final CV top-off adds some time in practice.)

Example 4: C-rate at 100 kW. 100 ÷ 61.44 ≈ **1.6C**. If the cell is rated 2C charge only above 20 °C, a cold pack will be limited well below 100 kW.

Example 5: regen from braking at 100 km/h. 1,700 kg car: ½ × 1,700 × 27.8² = 0.182 kWh of kinetic energy; at 65% regen efficiency, about **0.12 kWh** recovered per full stop from 100 km/h.

8. Why a 150 kW charger doesn't mean 150 kW

1. **Vehicle maximum:** the car may accept only 100 kW.
2. **Voltage × current limits:** a 400 V car at 30% SOC may be at about 360 V; with a 375 A charger cable limit, maximum = 135 kW.
3. **SOC:** taper above about 50–60%.
4. **Temperature:** a cold or hot pack is current-limited.
5. **Shared power:** many stations split one power cabinet between two plugs.
6. **Station or grid derating:** in hot weather or when the site's grid supply is limited.

9. V2X, swapping and standards

Bidirectional charging

Mode	What it does	Example
V2L (vehicle-to-load)	Powers appliances from the car's sockets	Hyundai Ioniq 5, Tata Curvv EV, MG ZS EV (check variant)
V2H (vehicle-to-home)	Backs up a home through a bidirectional charger	Nissan Leaf in Japan, some US systems
V2G (vehicle-to-grid)	Sells or shifts energy for the grid	Pilots in Europe, UK, US; trials discussed in India

Cost to the battery: extra cycling; most V2X use is shallow and slow, so ageing impact is modest but non-zero; warranties may limit it.

Battery swapping. A depleted pack is exchanged for a charged one in minutes. It decouples battery cost from vehicle price (battery-as-a-service) and suits high-utilisation two- and three-wheelers. Examples in India: SUN Mobility, Battery Smart, Gogoro's India operations, Honda's swappable packs. Challenges: pack standardisation across OEMs, ownership and warranty, capital in spare packs, safety and traceability. NITI Aayog published a draft battery swapping policy in 2022; check its current status.

Connectors and standards

Standard	Type	Where
Type 2 (IEC 62196-2)	AC	India, Europe; most Indian cars
CCS2 (Combo 2)	DC	India and Europe; standard for Indian cars
Bharat AC-001 / DC-001	AC / DC low power	Early Indian public charging (low voltage)
LECCS / IS 17017-2-6	AC/DC for light EVs	Indian two- and three-wheelers (adopted 2024; check rollout)
GB/T	AC / DC	China
CHAdeMO	DC	Japan; older Nissan Leaf
NACS (SAE J3400)	AC/DC	North America (Tesla standard, adopted by many OEMs)
ISO 15118	Communication	Plug & Charge, V2G

Grid integration. Fast-charging hubs need MW-scale connections, transformers and demand charges. Solutions include stationary storage buffers, managed (smart) charging, time-of-day tariffs, and solar.

10. Real-world EV examples

- **Indian compact EVs** typically accept about 50–100+ kW DC, depending on model and variant; check current specifications.
- **Hyundai E-GMP (800 V)** advertises 10–80% in about 18 minutes on a 350 kW charger.
- **BYD Super e-Platform (1,000 V)** announced in 2025 with megawatt-level charging claims; deployment is mainly in China.
- **Two- and three-wheeler fleets** in India rely heavily on swapping and slow AC charging.

11. Manufacturing applications

- **Charging-curve calibration:** cell suppliers deliver plating-safe current maps from three-electrode cell tests; the OEM validates on packs at multiple temperatures.
- **EOL charge-communication test:** each vehicle is tested for AC and DC charging handshakes.
- **Connector quality:** inlet contact resistance and temperature sensors are tested; poor contact causes hot inlets and charging derating.
- **OBC and DC-DC:** often integrated into one unit; thermal and EMC testing on the line.

12. Common mistakes and misconceptions

- **"Fast charging always destroys batteries."** Within the BMS's limits and temperature control, the effect is modest; frequent fast charging in heat or cold accelerates ageing more.
- **"Charging 0–100% is how to compare chargers."** 10–80% is the meaningful benchmark.
- **"kW on the charger = kW into the battery."** Losses and cooling loads reduce it.
- **"Regen fully charges the battery downhill."** Regen is limited at high SOC and low temperature.

13. Practical case study: slow charging complaints in winter in North India

Problem. Customers in Delhi report 25–30 kW charging on 60 kW chargers in December mornings.

Analysis. Pack temperatures of 8–12 °C after overnight parking; BMS map limits current to avoid plating; no navigation-based preconditioning on the model.

Solution. OTA update adding preconditioning when a charger is the destination; in-app charging advice; next model year adds a heat pump.

Lesson. Charging performance is a thermal and software problem as much as a hardware one.

14. Revision notes

- AC charging uses the onboard charger; DC uses the station's converter.
- Charging tapers because of plating risk, voltage limits, heat and imbalance.
- Use SOC bands with average power and efficiency to estimate time.

- 10–80% is the meaningful fast-charge benchmark.
- India: Type 2 AC and CCS2 DC for cars; LECCS for light EVs; swapping strong in 2W/3W.

15. Glossary

Term	Meaning
OBC	Onboard charger
CC-CV	Constant current, then constant voltage
Taper	Falling charge power as SOC rises
Plating	Metallic lithium deposited on the anode
Preconditioning	Warming or cooling the pack before charging
V2L / V2H / V2G	Vehicle-to-load / home / grid
BaaS	Battery-as-a-service
Plug & Charge	Automatic authentication via ISO 15118

16. Knowledge check

Conceptual

1. Why is AC charging power limited even when the grid supply is large?
2. Why does charging current fall above about 80% SOC?
3. How does low temperature increase plating risk?
4. Why does an 800 V architecture enable faster charging?
5. What are the business advantages of swapping for two-wheeler fleets?

Numerical

1. A 30 kWh pack charges 20–80% at an average 25 kW, 94% efficient. How long?
2. A 400 V pack at 350 V with a 200 A cable limit: maximum charging power?
3. A delivery scooter (150 kg with rider) brakes from 40 km/h to 0 fifty times a day at 60% regen efficiency. Energy recovered per day?

Application

1. An OEM advertises "up to 150 kW" charging. What data would you ask for to judge real-world performance?

2. A fleet operator chooses between depot AC charging and swapping for 500 three-wheelers. What factors would you compare?

17. Answer key

Conceptual

1. The onboard charger's power electronics are limited by cost, mass, packaging and cooling inside the car (commonly 7–22 kW).
2. The cells near their maximum voltage, and the anode's intercalation rate slows at high SOC, so current must fall to avoid plating and overvoltage.
3. Lithium diffusion and charge-transfer kinetics slow down, so lithium accumulates at the anode surface and deposits as metal.
4. For the same current limit of cables and connectors, power doubles with voltage; heat in cables (I^2R) stays the same.
5. Fast turnaround (minutes), vehicles sold without battery (lower upfront price), batteries managed centrally (charging, health, safety), and higher vehicle utilisation.

Numerical

1. 0.6 × 30 = 18 kWh → 18 ÷ (25 × 0.94) = 0.766 h = **46 min**.
2. 350 × 200 = **70 kW**.
3. v = 11.1 m/s; KE = ½ × 150 × 11.1^2 = 9,259 J; × 0.6 = 5,556 J; × 50 = 277,800 J = **0.077 kWh/day** (about 2–3 km of scooter range).

Application

1. The full charging curve (kW vs SOC) at 25 °C and at low and high temperatures; 10–80% time; maximum current and voltage at each SOC; whether preconditioning exists; performance on 60 kW chargers common in India; degradation data with frequent fast charging.
2. Upfront capex (chargers vs spare packs and stations), vehicle downtime and utilisation, electricity tariff and demand charges, land and grid availability, battery ownership and warranty, battery life under each regime, safety and monitoring, and interoperability with OEMs.

18. Further learning

- IEC 61851 and IEC 62196 (charging system and connectors); ISO 15118.
- Ministry of Power (India), *Charging Infrastructure for Electric Vehicles: Guidelines and Standards* (latest revision).
- NITI Aayog, *Draft Battery Swapping Policy* (2022).

- Waldmann et al., "Temperature dependent ageing mechanisms in Li-ion batteries," *J. Power Sources* 2014.

Next: Module 10, degradation, ageing and second life.